Cybersecurity Projects (TuteDude)

1. Comprehensive Cyber Attack Simulation

Objective:

Understand how a complete cyber attack works from initial discovery to data theft and how to defend against it.

What You'll Do:

You will create both attacker and victim machines in a safe lab environment. Step by step, you'll perform reconnaissance, exploitation, privilege escalation, and data exfiltration. Then you'll switch to the defender's role: analyze what happened, apply security fixes, and re-test the system.

This project helps you understand the full attack lifecycle and how security controls improve a system's posture.

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2. Advanced Phishing Attack Scenario

Objective:

Learn how phishing attacks succeed and how organizations detect and prevent them.

What You'll Do:

You'll simulate a phishing campaign where fake emails are used to trick users into clicking links or downloading payloads. After gaining controlled access to a test system, you'll practice basic post-compromise activity and safe data exfiltration.

You'll also document Indicators of Compromise (IOCs) and recommend protection measures. This builds strong awareness of social engineering risks.

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3. Building a Virtual SOC Environment

Objective:

Experience how a real Security Operations Center (SOC) works.

What You'll Do:

You'll build a virtual SOC using Splunk for log analysis and Wazuh for endpoint monitoring. You'll ingest logs, create dashboards, analyze alerts, correlate events, and perform basic incident response.

This project teaches daily SOC operations and how teams detect and respond to threats.

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4. Forensic Investigation of Cyber Attacks (Memory Forensics)

Objective:

Recover evidence from compromised systems.

What You'll Do:

You'll analyze memory dumps from infected machines to find malicious processes, network connections, and other volatile artifacts stored in RAM. You'll build timelines and summarize findings.

This shows how investigators uncover hidden evidence during incident response.

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5. Network Packet Analysis

Objective:

Learn how to spot attacks by examining network traffic.

What You'll Do:

Using Wireshark, you'll capture and inspect packets, identify suspicious patterns, and investigate issues such as abnormal connections or potential denial-of-service activity.

You'll learn how raw network data reveals security problems.

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6. ICS / SCADA Security Testbed Setup

Objective:

Understand security in industrial and critical infrastructure environments.

What You'll Do:

You'll set up a simulated ICS/SCADA lab and study its unique challenges, including legacy systems and real-time operations. You'll perform basic threat modeling and apply security controls such as segmentation and monitoring.

This project introduces Operational Technology (OT) security concepts.

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7. Honeypot Deployment

Objective:

Observe attacker behavior in real time.

What You'll Do:

You'll deploy a honeypot (a decoy system) to attract attackers. By monitoring activity, you'll collect IPs, commands, and techniques used by intruders.

You'll summarize attacker behavior and suggest defensive improvements based on what you observe.

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8. Retrieval of Lost Data in Investigations

Objective:

Recover deleted or missing digital evidence.

What You'll Do:

You'll use forensic techniques to restore deleted files, emails, and artifacts from storage media while maintaining investigation integrity.

This teaches the importance of data recovery in digital forensics.

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9. Incident Investigation Using the Splunk BOT Dataset

Objective:

Practice real-world log-based incident investigations.

What You'll Do:

You’ll analyze the Splunk BOT dataset, build search queries, identify suspicious patterns, and reconstruct multi-stage attacks from logs.

This project strengthens your ability to detect breaches through log correlation and analysis.